

Complexity (discrete elements) $\rightarrow$ I.C

I.C

VS

D.C

- monolithic silicon
- on a same silicon chip.

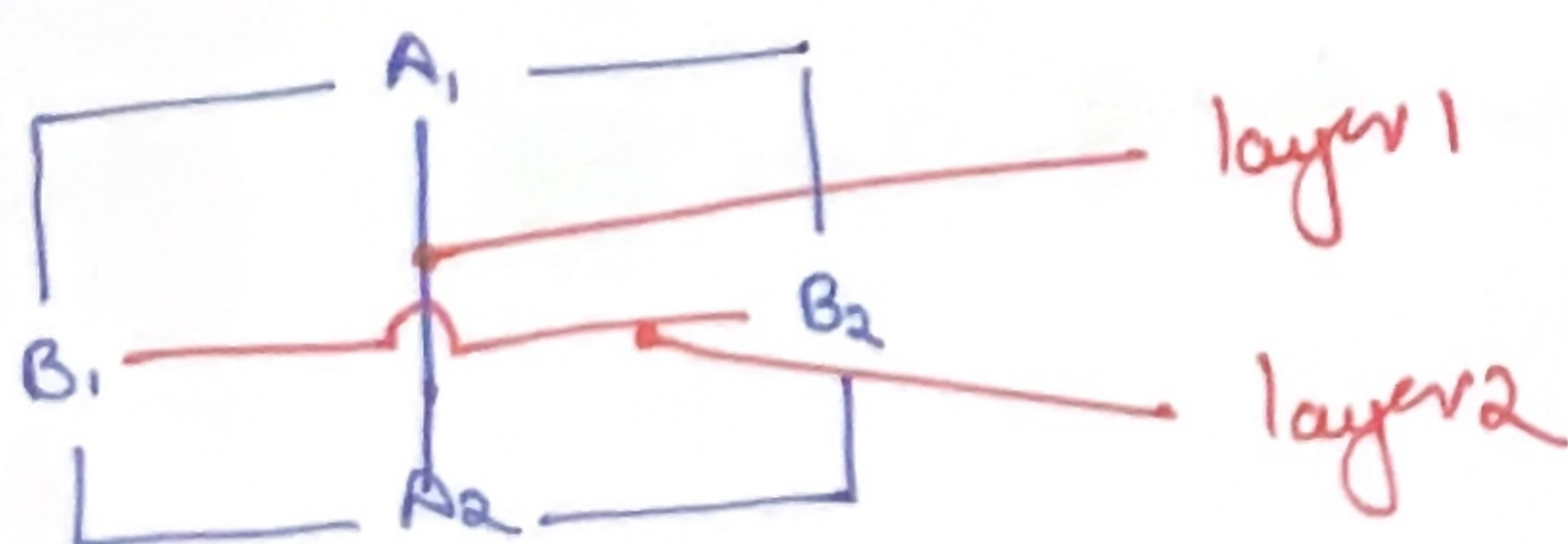
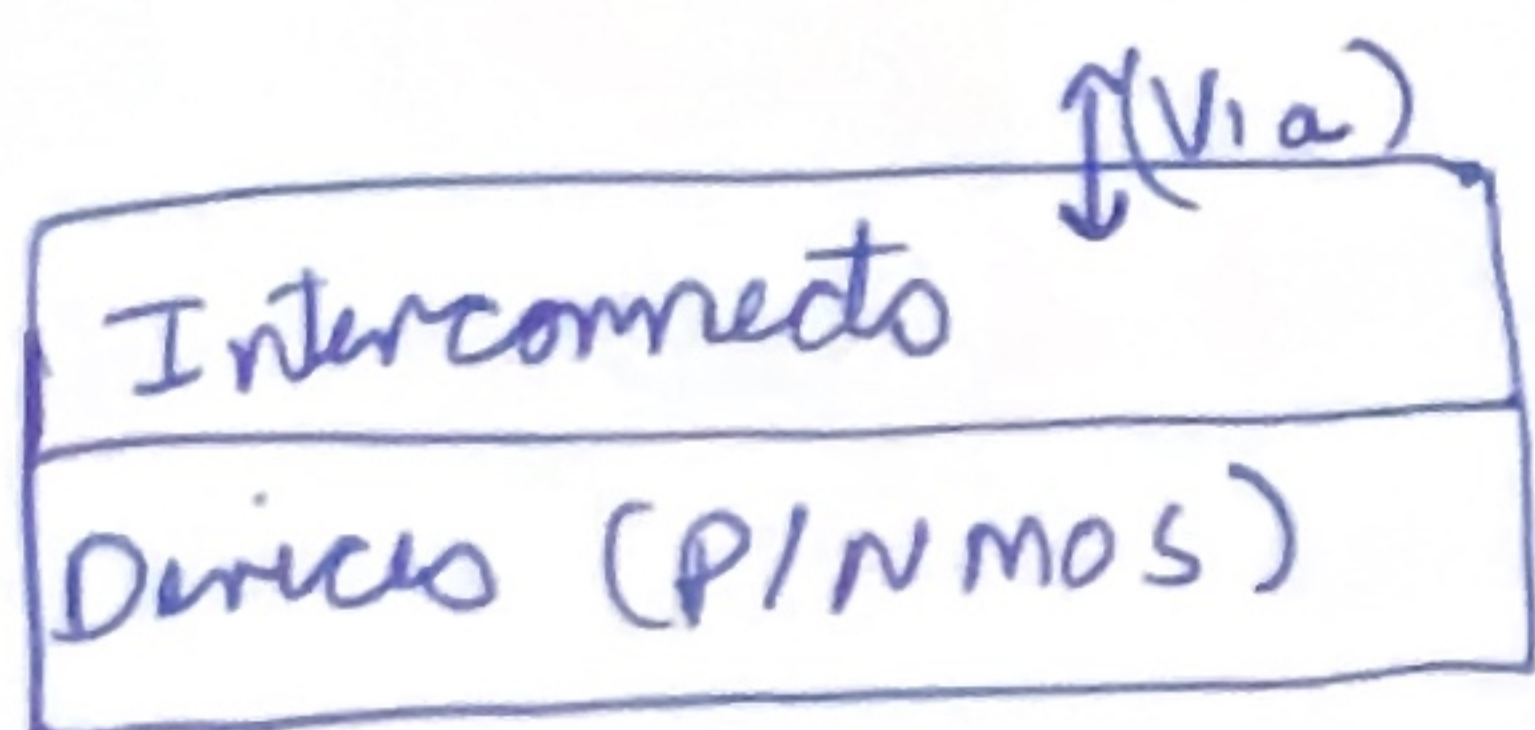
Photolithography - IC Tech - Complex Circuit.

Moore's Prediction. No of components in an IC will double each year, later every 2 years.

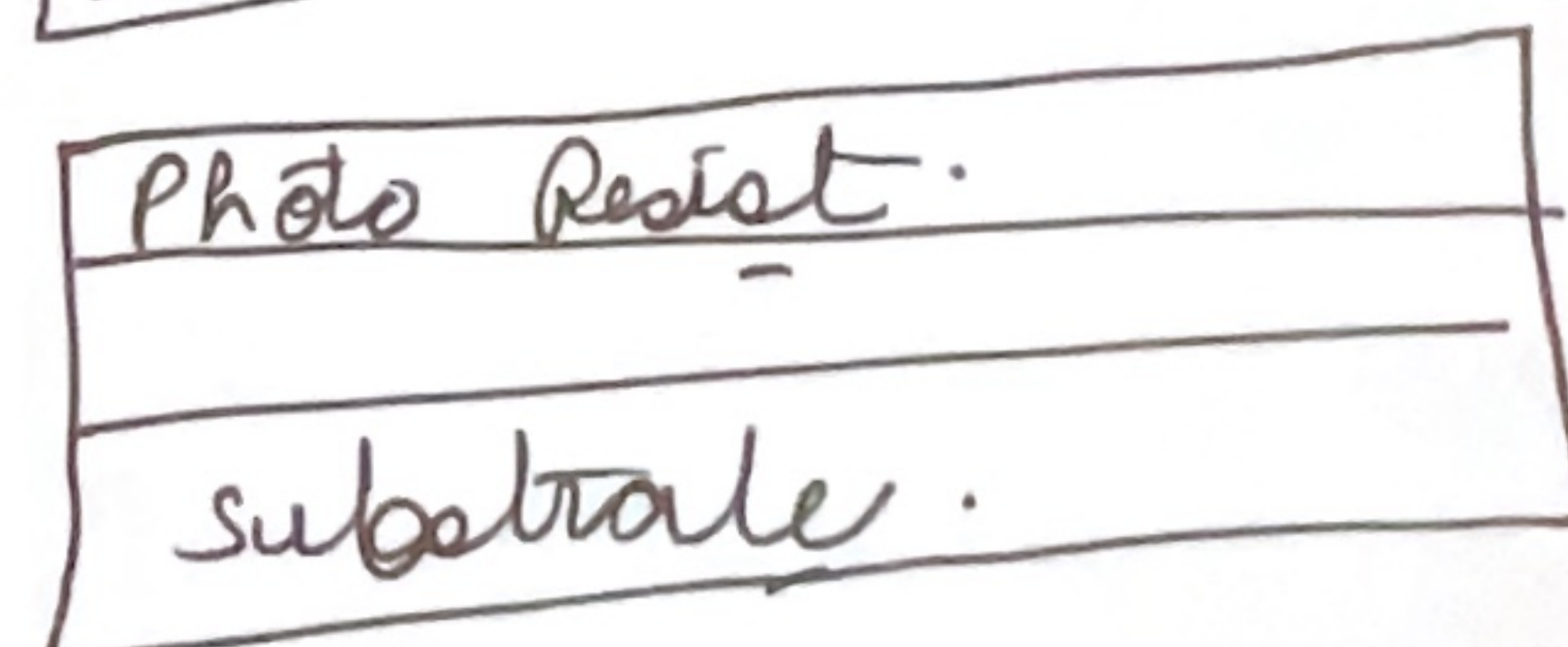
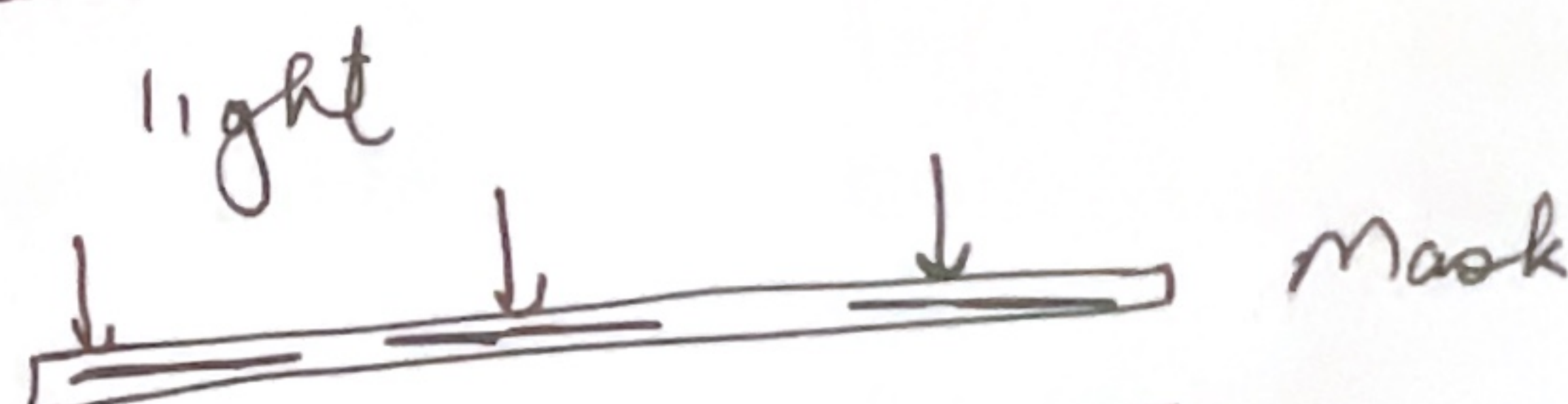
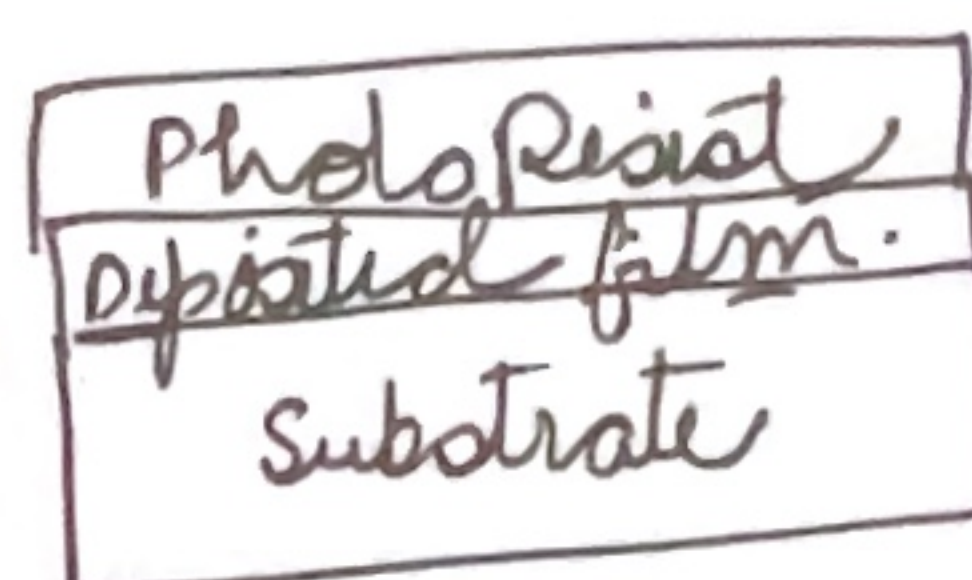
$\rightarrow$ size of Transistors - speed, energy efficiency, cost per transistor.

$\rightarrow$ structure of IC
features of IC

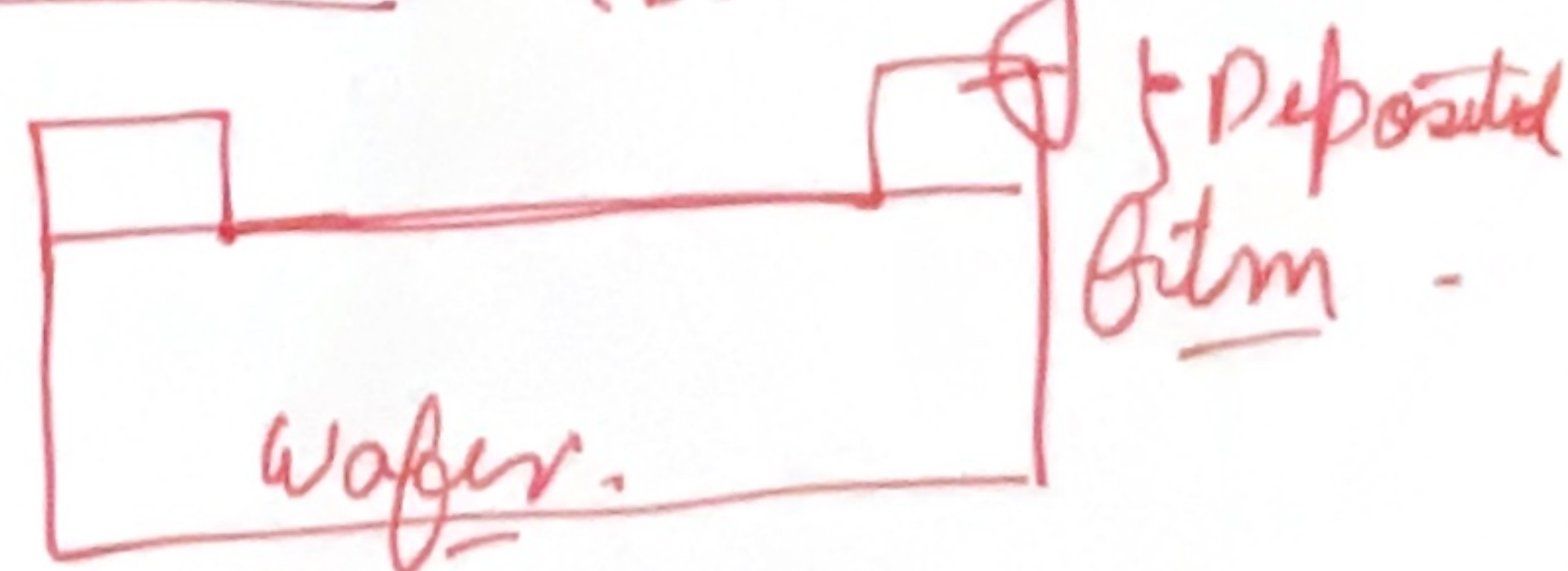
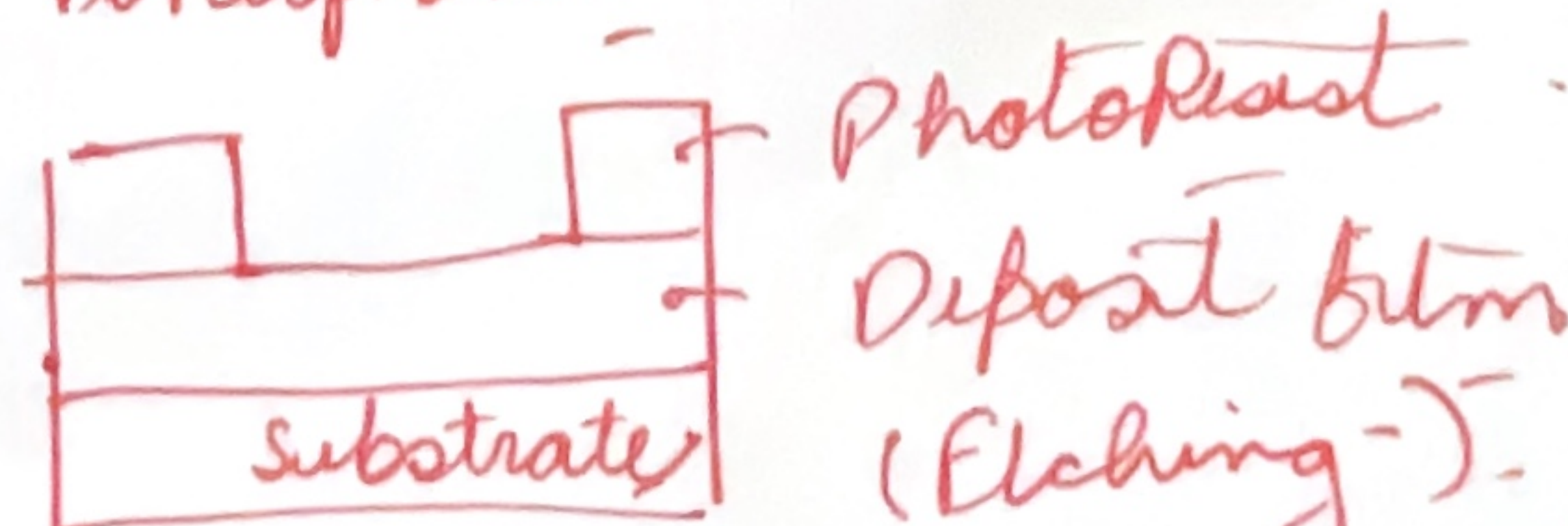
$\rightarrow$ Substrate



$\rightarrow$ Connection inside this box without short.
(Wire to Run in layers)
Different Planes



$\rightarrow$ Development



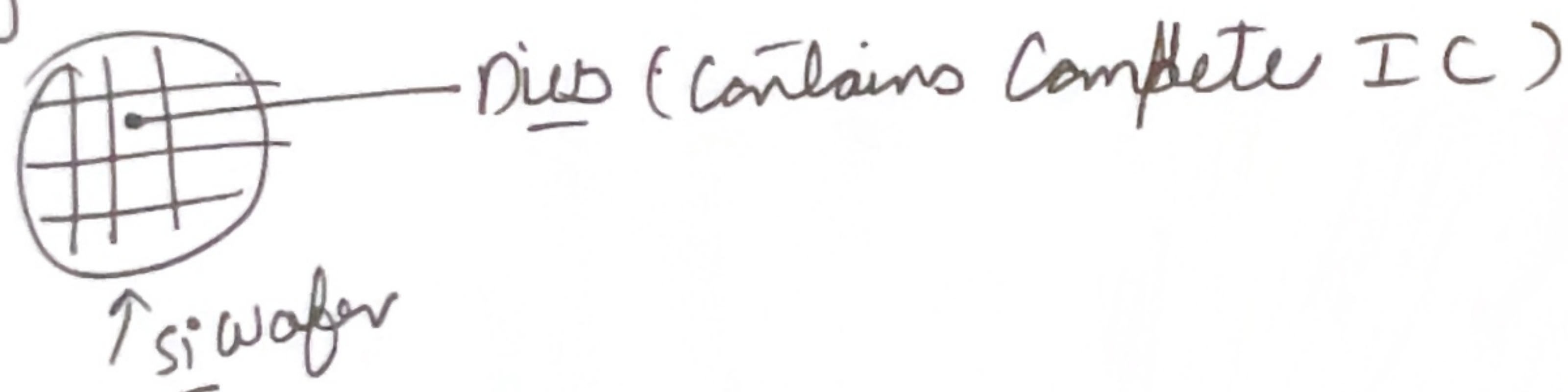
- layer's defined by mask & fabricated using photolithography.
- Transferring geometric shapes that are defined on a mask, to the surface of Si wafer.

What is a Mask.

Wafer - Thin slice of silicon that serves as a substrate for an IC.

- Ingots - Massive cylindrical single crystal of Si.
 - CZ process
 - Pure seed crystal + Pulling it up, (1425°C)
 - Ingots → wafer →

- single Si wafer → multiple I.C → Dies.



Dies sliced out of wafer of Si.

How we can make good die / defect free die.

Die → Can also contain large no of transistors.

→ encapsulated inside package (dies).

Designing	VS	Fabrication
Parameter + Composition of a circuit To achieve desired functionality.		Actual creation of IC for a given design.

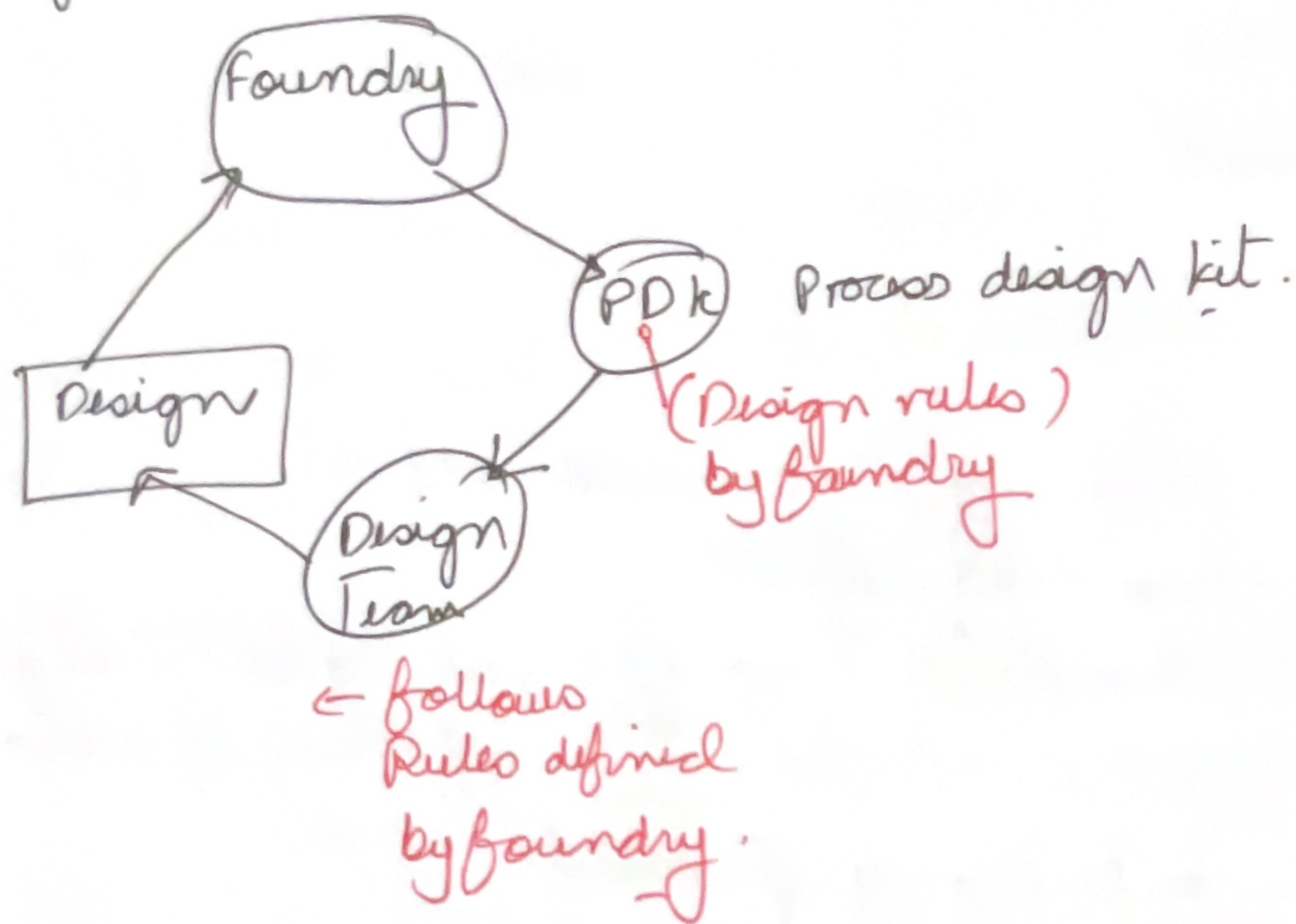
Fabrication: S.C. foundries.

- clean Room + investment
- shelf life (limited span of time).
- Full potential usage.

Business Model

- Fabless design Companies.
- Merchant Foundries → only fabrications (for others)
 - TSMC, UMC, GF etc.
- Integrated Device Manufacturers (Design + fabrication)
 - Intel + Samsung.

Design & Fabrication - Related →



Concepts of IC-II

VLSI design flow → To design IC such that it delivers the Required functionality and behaviour.

- Types of IC.

• scope of Application - ASIC

- General Purpose.

ASIC

general Purpose.

IC for digital camera.
Audio, Video, Processors.

MP, memory, FPGA

not Programmable
for wide variety of
different tasks

non Programmable

low

more production.

Design styles.

(3)

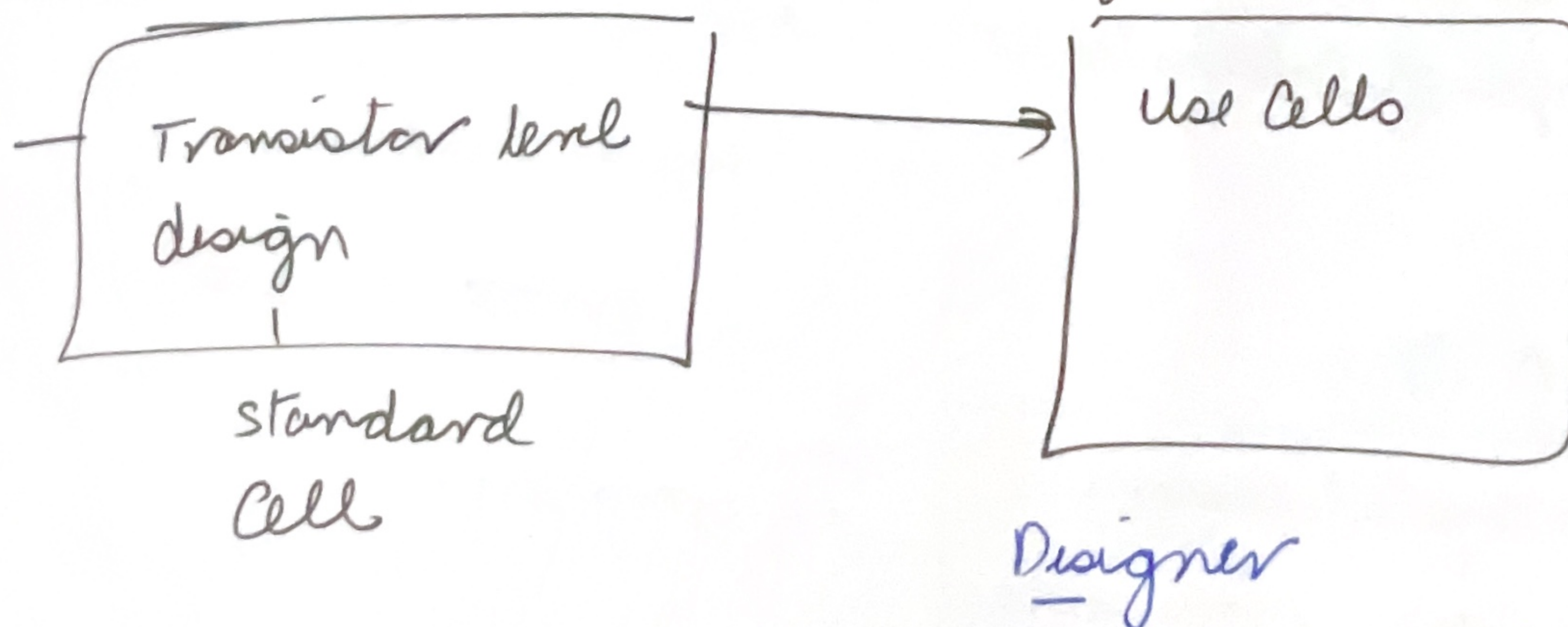
- Full custom
- Standard cell based
- Gate array based
- FPGA based

Full custom design :

- Designing at a transistor level.
- huge design effort.
(Analog ckt design / Mixed Signal Design)
- HP (some portion) To improve performance
- Can be highly optimized.

Standard cell → We design - logic gates, are made on ~~transistor~~ transistor level.
(and, or gates) and put them in a library.

→ Cells as a unit, and use in design



- Rows of standard cells, standard cells can be changed.

Gate array - Transistors are predefined (Pre Placed)
- Designer can change the interconnections b/w

transistors:

→ Top most layer (i.e.) layers of interconnections are design specific.

FPGA based design.

→ Hardware is fixed

IC h/w is fixed

→ Program the Connection b/w Hardware.

- Array of logic blocks, I/O blocks, Routing channel

- MP, Blocks for DSP

- Design styles.

(Diff b/w FPGA & cell based design)

→ Gate array based

Custom Design	standard All Based Design	Gate Array based	FPGA based design.
Design specific customization at the level of transistors and layout	standard cells already made by transistor level design interconnections	only Top level interConnects are design specific	Hardware is fixed. only change Connection + logic.
High effort	high effort	Top few layers design specific.	None.
Best,	good		Performance Power Area

What To choose -

economics - Fixed Cost

Variable Cost

- Design effort
- Software, hardware cost.
- Cost of Masks, (fabrication) (Depends on no of layer)

- Cost of Wafer
- Cost of die
 - size of die.
 - Yield (Higher no of good / defect free die).

standard cell based

FPGA based

Fixed Cost

↑ high designing Cost, Tools, Mask.

↓ low tools for programming

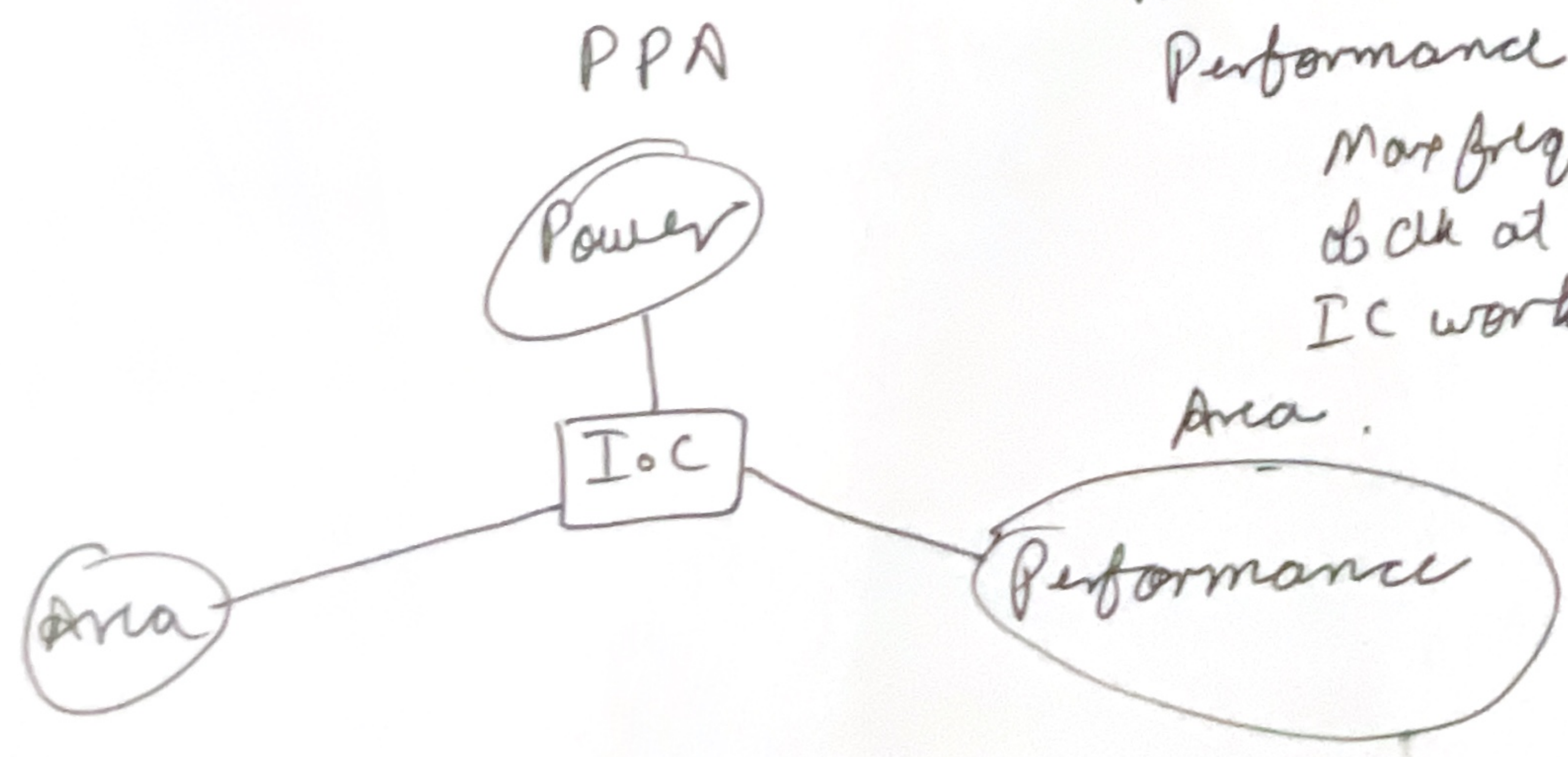
Variable Cost / Unit Product

↓ Die area ↓ more dies on a wafer

High Cost (if we didn't utilize all complete hardware) large die size, low yield.

Figures of Merit - How To Measure goodness of chit.

Power - Static + dynamic
Performance - Max frequency of clk at which IC works



Figures of Merit

- Testability - how easily can we test the product after manufacture.

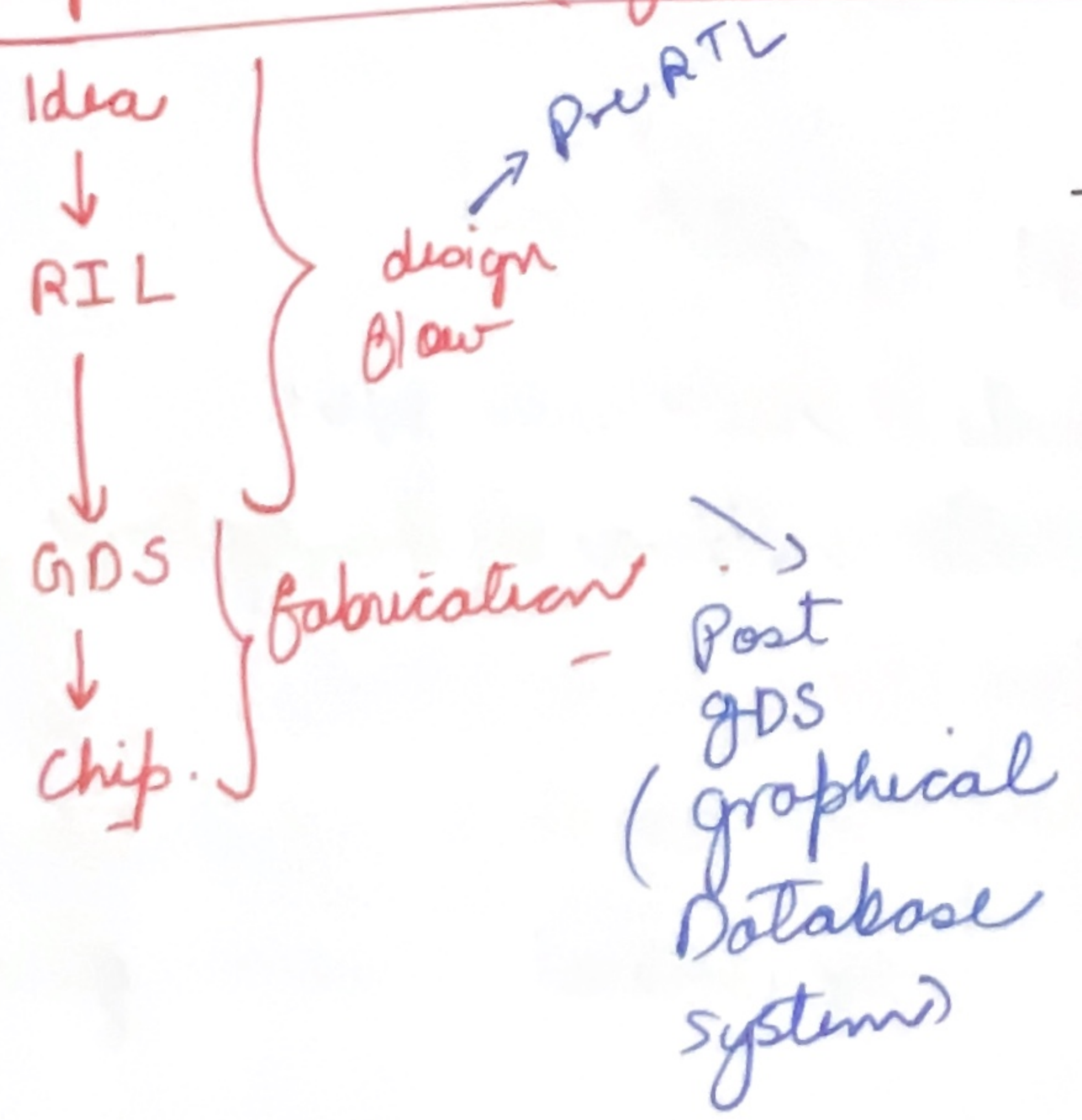
Reliability $\Rightarrow$ Quality of chip, should not degrade

Schedule - How much time we take to design that chip.
(TTM) Time To Market.

QoR (Quality of Result) - Trade off

Acceptable Form (Figure of Merit)

Abstraction
hiding lower level details



- Why to abstract.
- optimization

- To find Right combination of design parameters, To obtain desired QoR

Turn Around Time $\rightarrow$
Time Taken To make change in a design.

$\rightarrow$ To explore design details

RTL $\rightarrow$ Very high abstraction $\rightarrow$ Turn Around Time



Increased Graphical Database system.

GDS

Abstraction

- formula $F = (A + B)'$ - high Abstractions low turn around time
- using AND/NOR gate. - greater Accuracy in evaluation.

PRE-RTL methodologies (system level design)

Idea - Market Requirement
financial viability.
specification $\rightarrow$ features -

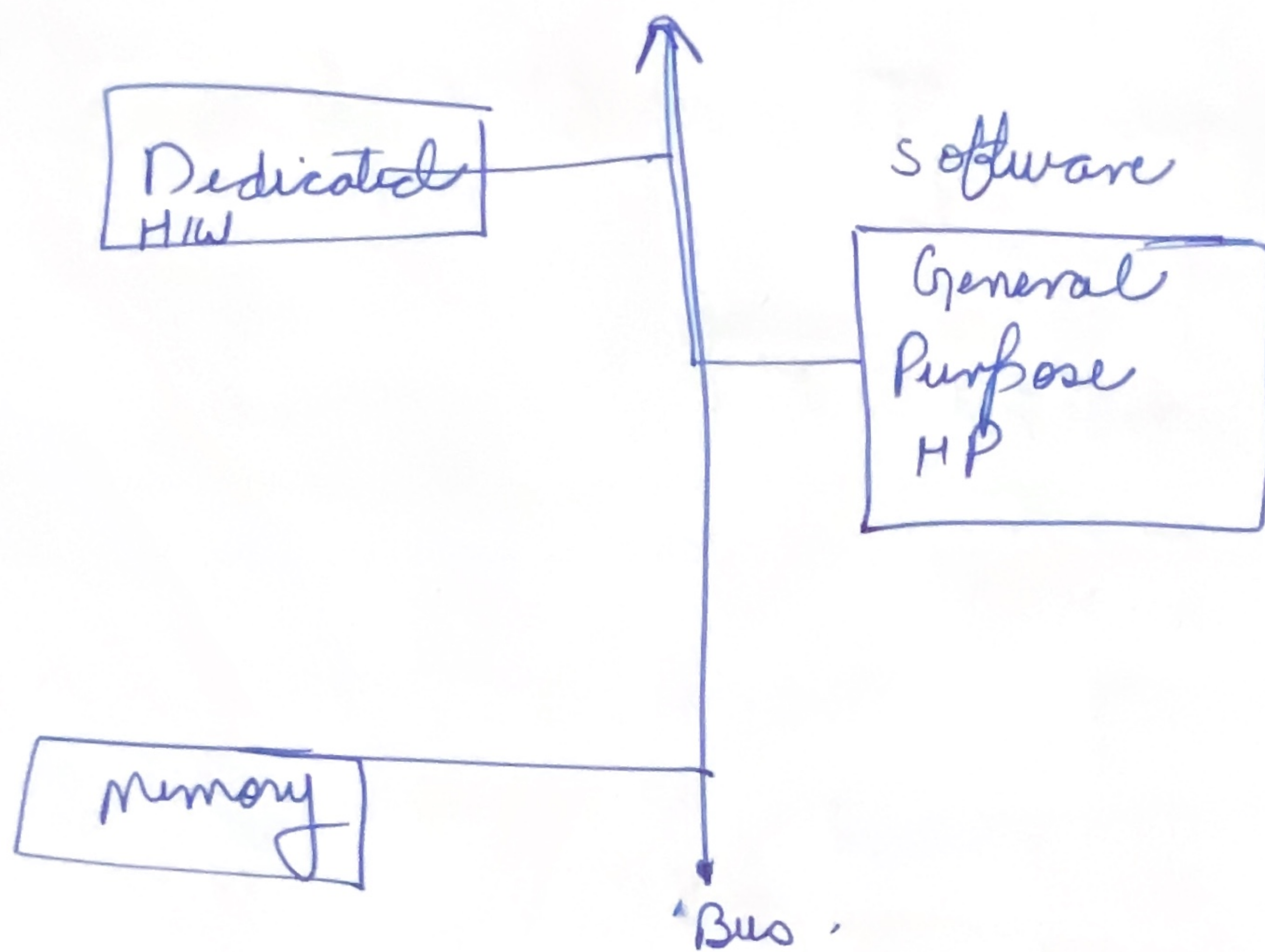
PPA (figures of merit)

TTM (schedule).

H-W / S-W Partitioning - Benefit of H/W & S/W

- Components implemented in H/W or S/W.
- H/W & S/W are both available, then we do system integration.

H/W	S/W
high speed/performance (Parallelism)	sequential manner operation
may need to change the H/W.	easy development less risk involved.
	easy to customize.



H/W Runs on parallel Ckts, and have a good PPA.

- Implemented in a full custom IC, ASIC, FPGA.
- Software → Run sequentially on a general purpose processor.

Video Compression.

DCT (Discrete Cosine Transform)
 (Hardware)
 (energy efficient customization)
 Frame handling & other Computation

Algorithm in software → Subset of fun's to h/w.

S, H, P (Acceptable Performance)
 N → Max No of functions to be Moved To Hardware in
 each iteration -

evaluate(H, S)

9

if (m, s) > P ——— already satisfied.

Profile
(H, S)

[$\beta_1 \rightarrow 10\%$ of time, $\beta_2 \rightarrow 20\%$ of time —]

↓
loop ($i = 1$ to N)

$S = S - \beta_i$
 $H = H \cup \beta_i$

↑
if not (Canl)
speed = new speed.

Bottle Neck fxn To the hardware.

- Iteration terminate
- if we moved all N fxn from (Sow To How) — Communication b/w S/W & H/W.
- or data dependency of H/W.

challenges

→ Performance estimation.

→ Verification on FPGAs block

→ Verification of HW & SW — Co-simulation.

PreRTL (System level Design)

What is library/instance pin.
pin A & I1

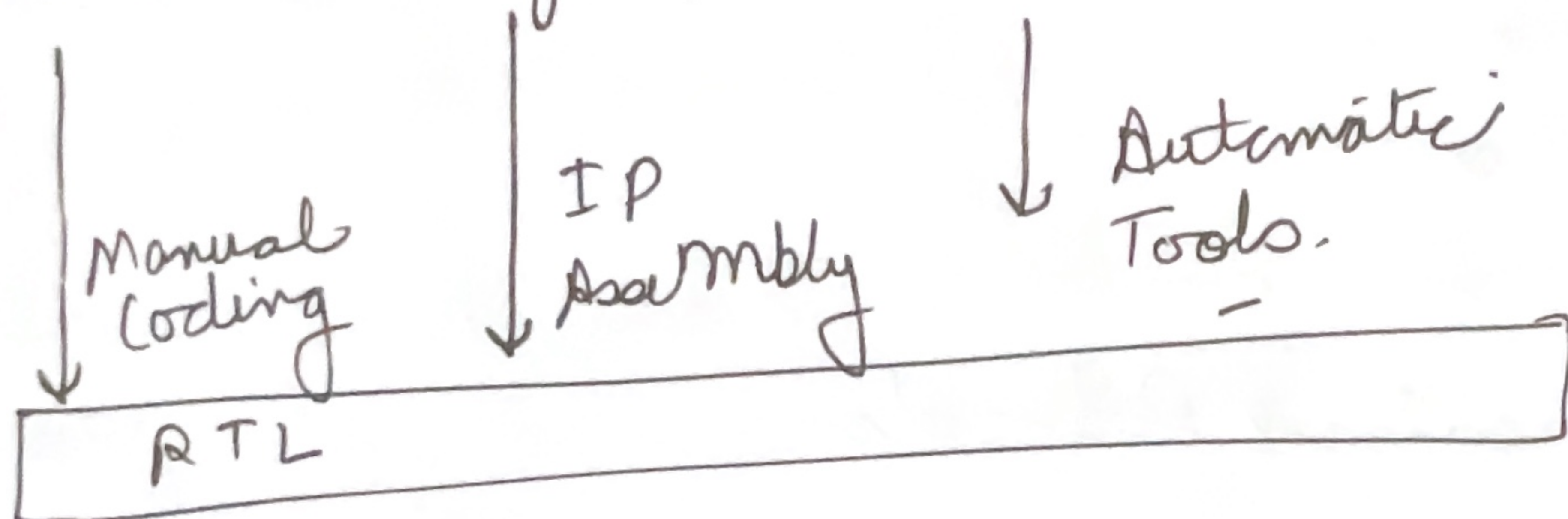
RTL

RTL TO GDS - Part 4

H/W - functional specification (C, C++) } implementation gap.

↓
RTL (Modelling of flow of data)

Functional specification (C, C++, system C) --



Reusing RTL → Popular on SOC.

- SOC can consists of Processors, Memory, peripherals, analog components, RF devices

- We Reuse existing components. } SOC.
- Lower Cost.
- Increases features

IP - Pre designed & Pre verified sub system.

- HW, SW, Verification

↓
HP → RTOS IP.

memory

→ How To share the information, How To Package the information, Uniform standards should be followed.

- instantiating of IPs in an SOC, and making their connections.

- Methodo - Metadata → Top level IP Modules, bus interfaces, ports, registers, and Required Configuration.

Metadata examples.

Generate Tools - To Create SoC level RTL with instantiated IP.

- Can also produce a verification environment and low level software drivers.

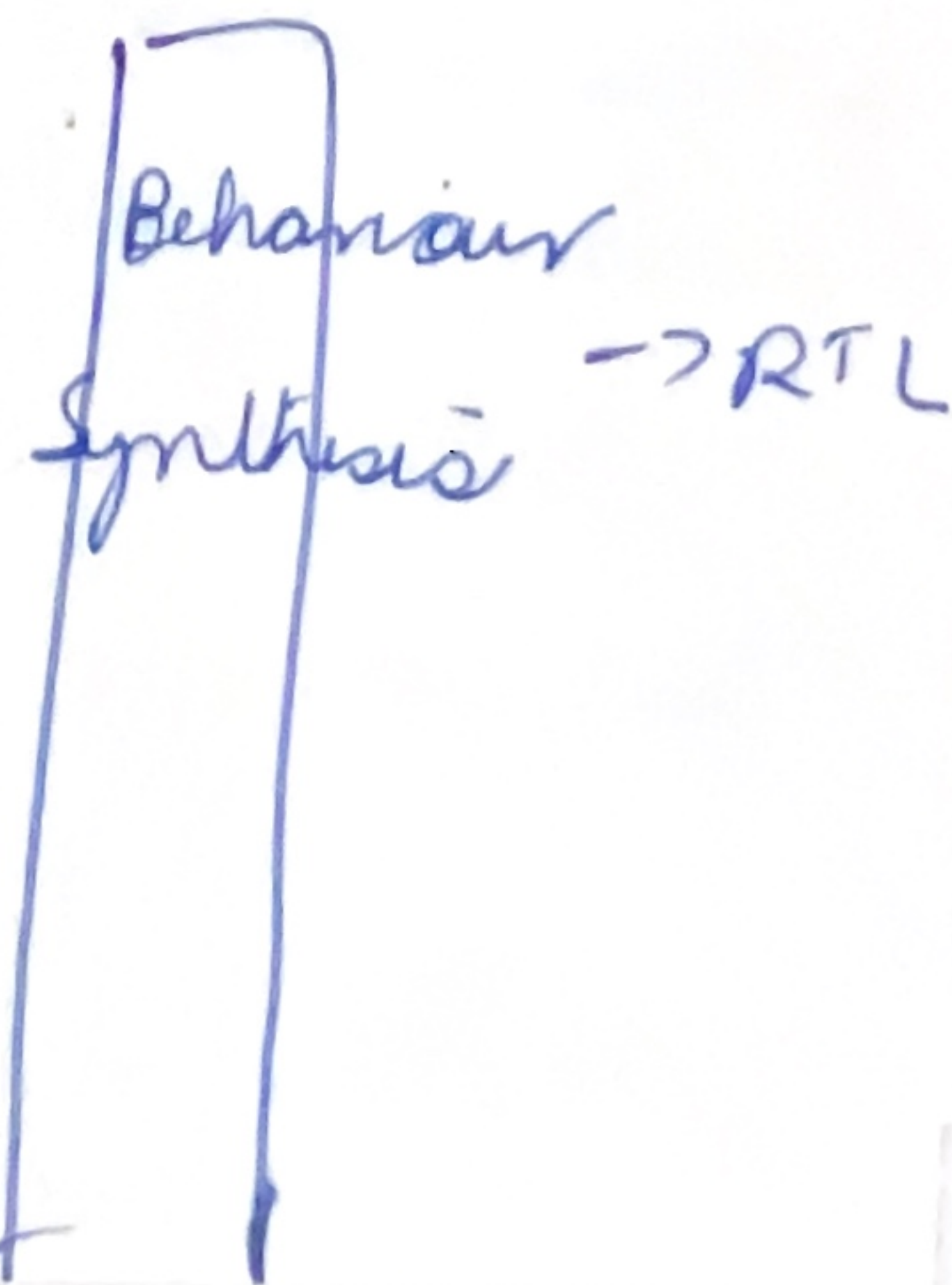
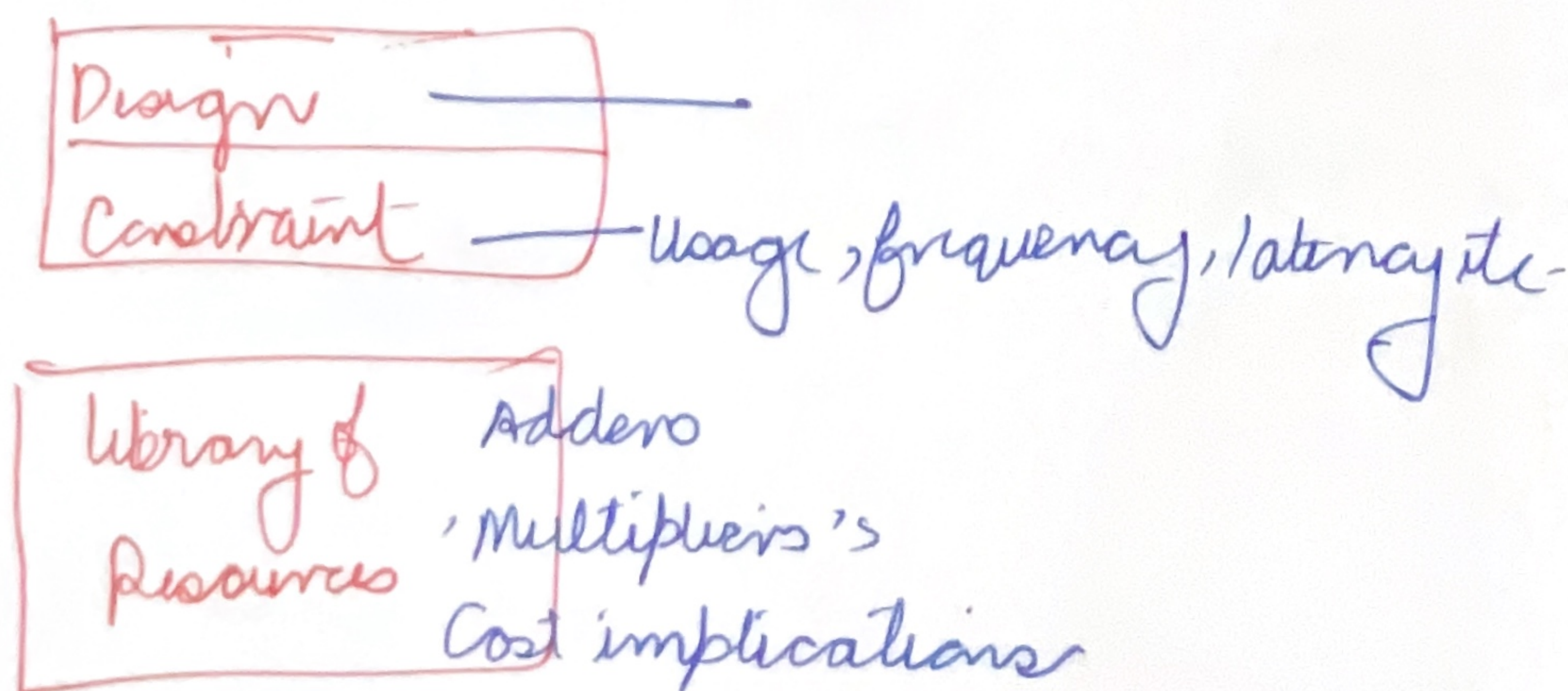
- Challenges
 - Bus widths
 - Power Modes
 - Communication Protocols.

How To Connect IP's

- Ad hoc bus-based
- structured Network Approach (NoC)
- Verification Challenges.
 - Huge functional space
 - SW & HW

- Functionality to RTL.

Non timed Algorithm $\rightarrow$ full timed Algo.



Behavioral Synthesis

- An untimed algorithm can be implemented in many ways.
- > Different implementation in different ways.
 - & different Cost Metrics.

Area: No of elements

Latency: No of clk cycles required before results are available

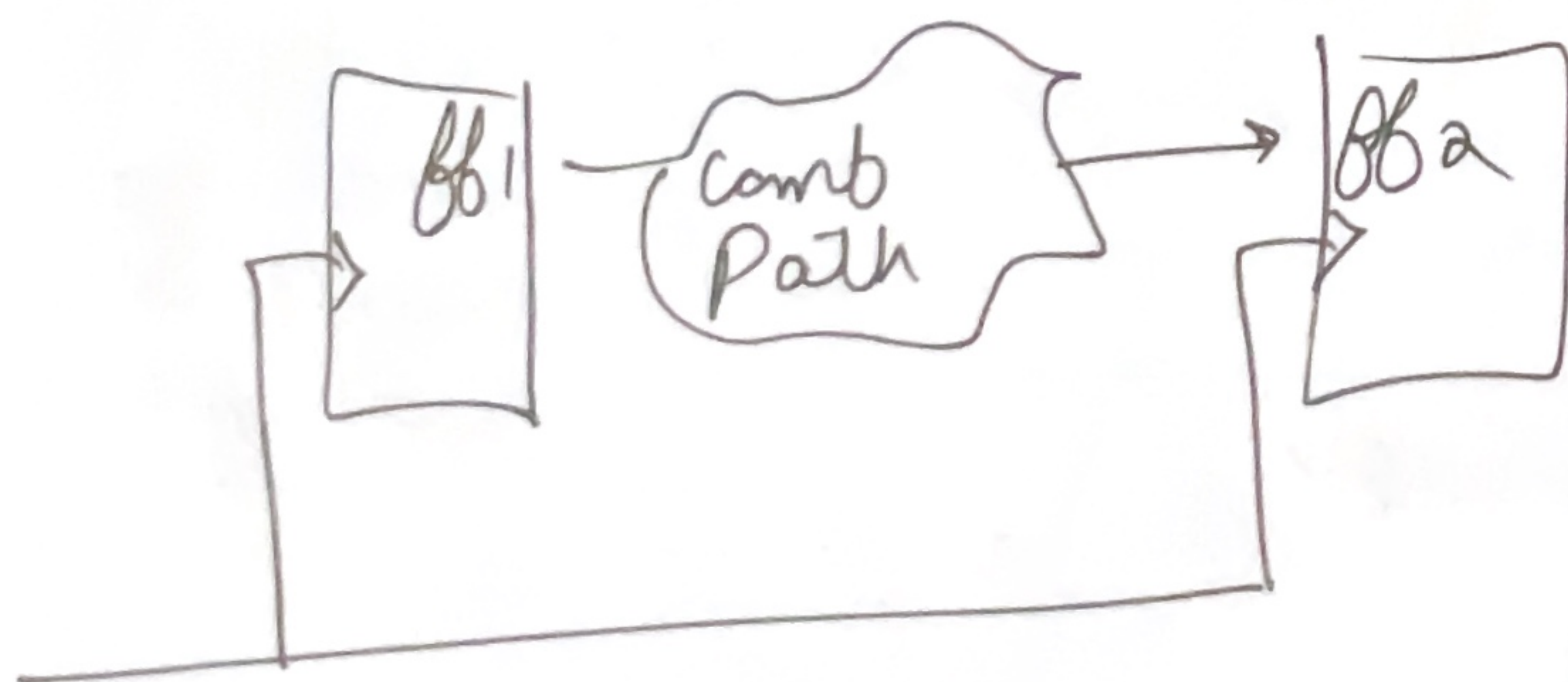
Max Clock: worst case Comb delay
Frequency

Max Clock Frequency

Path -> sequence of pins through which signals can propagate.

Combinational -> (No sequential elements)

Sequential elements -> FF / latches.



clk.

data launched must be captured by sequentially adjacent FF in the next clock cycle.

sequentially adjacent
FF if O/P of 1 FF is fed
as an I/P to other FF,
through a Combinational
Path.

$$T_{req} = T_{clk} +$$

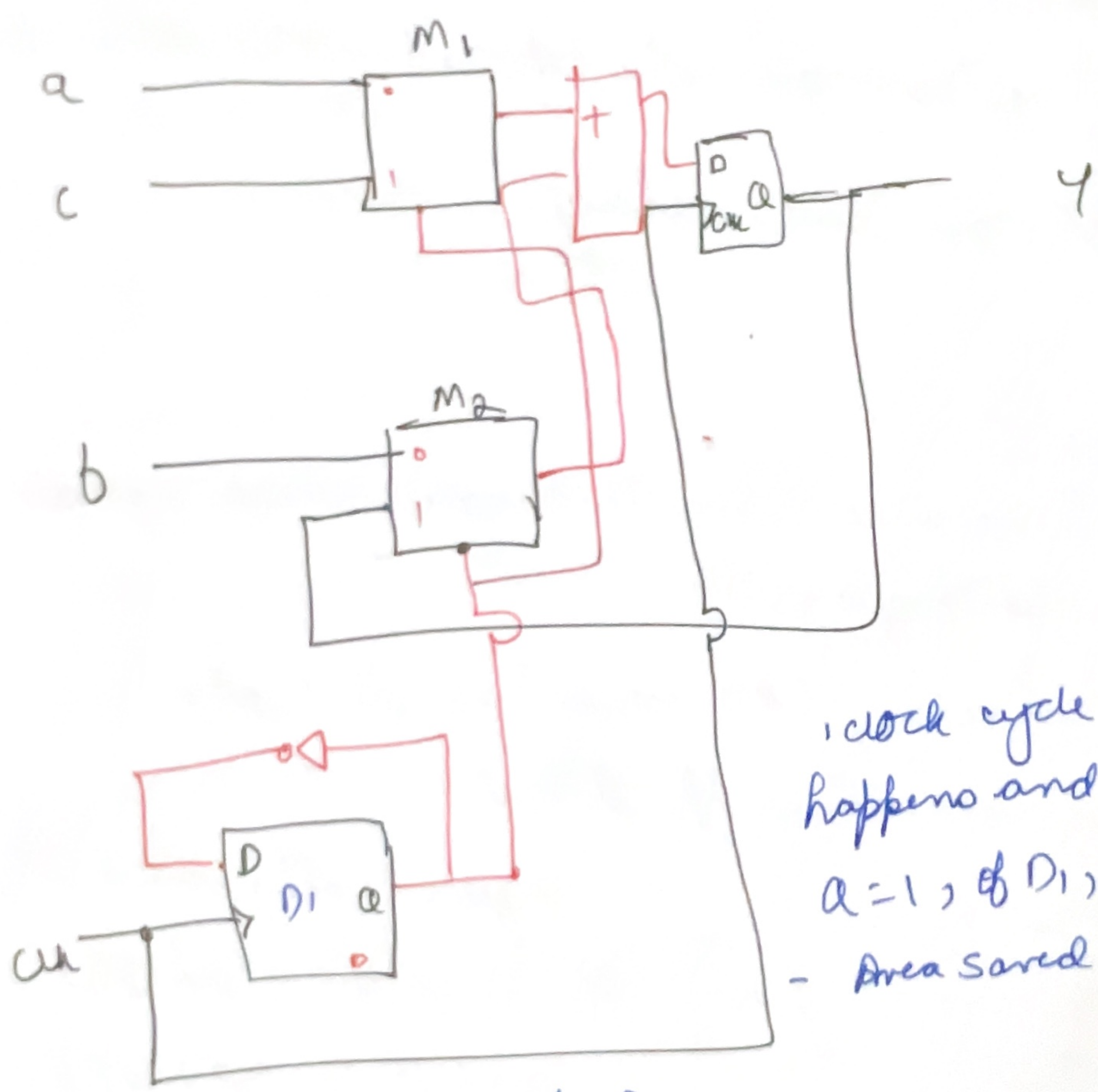
$$T_{arrival} = T_{clk} - q + T_{delay}$$

-> Path deciding t_{max} is
max Path, of Combination
of delay.

$y = a + b + c$

Cost Matrix $\rightarrow$ Ckt element used $\rightarrow$

latency - After how many clock cycles, will result be available.
Max delay of Comb Path $\rightarrow$



1 clock cycle, $a = 0$, i.e. $a + b$ happens and then,
 $a = 1$, of D_1 , i.e. $(a + b) + c$ happens
- Area saved $\uparrow$, latency $\uparrow$.
Producing result in 2 clk cycles.

- Delay $\uparrow$
Down side

Area - minimize		Through Put, latency -
Area $\downarrow$	mux, off one	

latency $\downarrow$
clk frequency $\rightarrow$ one with minimum delay.

behavioural synthesis Tool.

PreRTL (System level Design)

RTL

| (logic synthesis)

Netlist

What is library pin / instance pin.

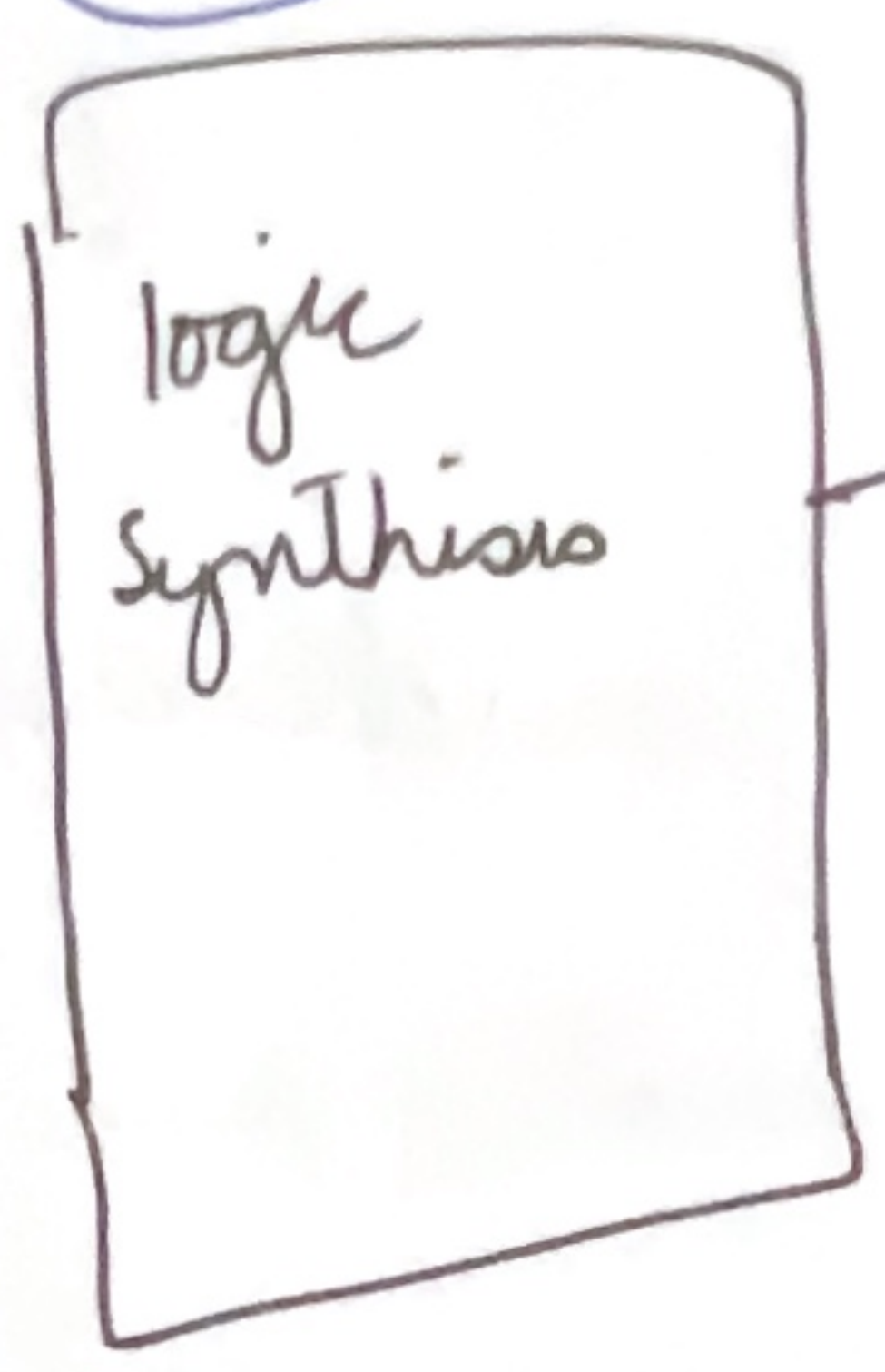
Pin A of I1

I1/A

Design / RTL

Library standard cells and macros

Constraint Timing, frequency, rise/fall time, SDC, synopsys design (constraint)



Netlist

interconnection of logic gates, same functionality as RTL, and Netlist is interconnection of logic gates.

- lib
- sdc
- v (netlist).

Netlist - schematic

RTL

logic synthesis

library mux, nbb → will be instantiated from lib, based on constraints, so lib may have multiple mux, nbb based on requirements

Constraint

(To Take The cells from library.)

Power Performance Area
→ PPA

Netlist terminologies:

Design - Top level entity that represents the circuit.

Ports - interfaces, of the design through which it communicates with internal world.

library → cells

cell (AN2)

Pin(A) —

Pin(B) —

communication with → the other components



Metadata examples.

Logic Synthesis Tasks:

RTL synthesis (RTL to generic logic gates) like simple elements.
→ i.e. well defined boolean fxn.

- RTL Construct to generic logic gates.
(is there any defined generic logic gates)

Netlist (generic gates) } logic optimization.

optimized Netlist (generally Area) } Technology Mapping.

- generic logic gates to standard cells.

- in given technology library.

- inverter with multiple

Netlist (Standard Cell)

Technology dependent

optimization

- drive strength.

optimized Netlist

└ Power, timing etc Calculations can be Done.

Document $\rightarrow$ Placement of standard cells, in Rows

(18)

Physical design

(16)

$\rightarrow$ Design in the form of netlist is converted to an equivalent design in the form of layout, or GDS (

Netlist

Design after logic synthesis

- Standard cells
- Tech library used during logic synthesis

Library

Constraints

Floor Plan

$\rightarrow$ GDS layout

[Placement
Connections
(Routing) To make

$\rightarrow$ Information on Row standard cells looks like in dimension

$\rightarrow$ LEF (Library exchange format)

- bounding box of standard cell, pin location, resistance technology specific).

• Size of layout we are trying

Metadata examples.

elements 15 17

Chip Planning

- floor plan
- PDN (Power delivery Network)

Placement → Placed Cells

CTS → clock signals, to make structure such that, the
all signal is delivered to all the entities in the design
that operate on clock

Global Routing

Detailed Routing

ECO

wire GDS →

mask layer

Making
mask

Chip Planning → Planning stage for layout.

→ major decision for layout

→ Partition the design into subsystem, and then

Combine,

→ location of bigger blocks are decided.

→ Decide location of standard cells, ok these are the
location of standard shape.

I, L

can decide the shape,

→ Also place I/O cells.

→ Deliver Power to layout, → Power Planning.

→ Congestion is also to be Taken Care of.

Placement → Placement of standard cells, in Rows

→ Macro's →

→ No of Macro's are low.

→ No of standard cells are multiple.

→ Placement of Macro's, ~~is~~ and standard cells is automated.

- Wire length minimization is also of concern (PDN) ~~is~~

CTS - Topology of clock n/w

- How clk reaches from clock source to FF's.

- highest Priority

- To minimize skew., (CTS Tool choose the Topology)

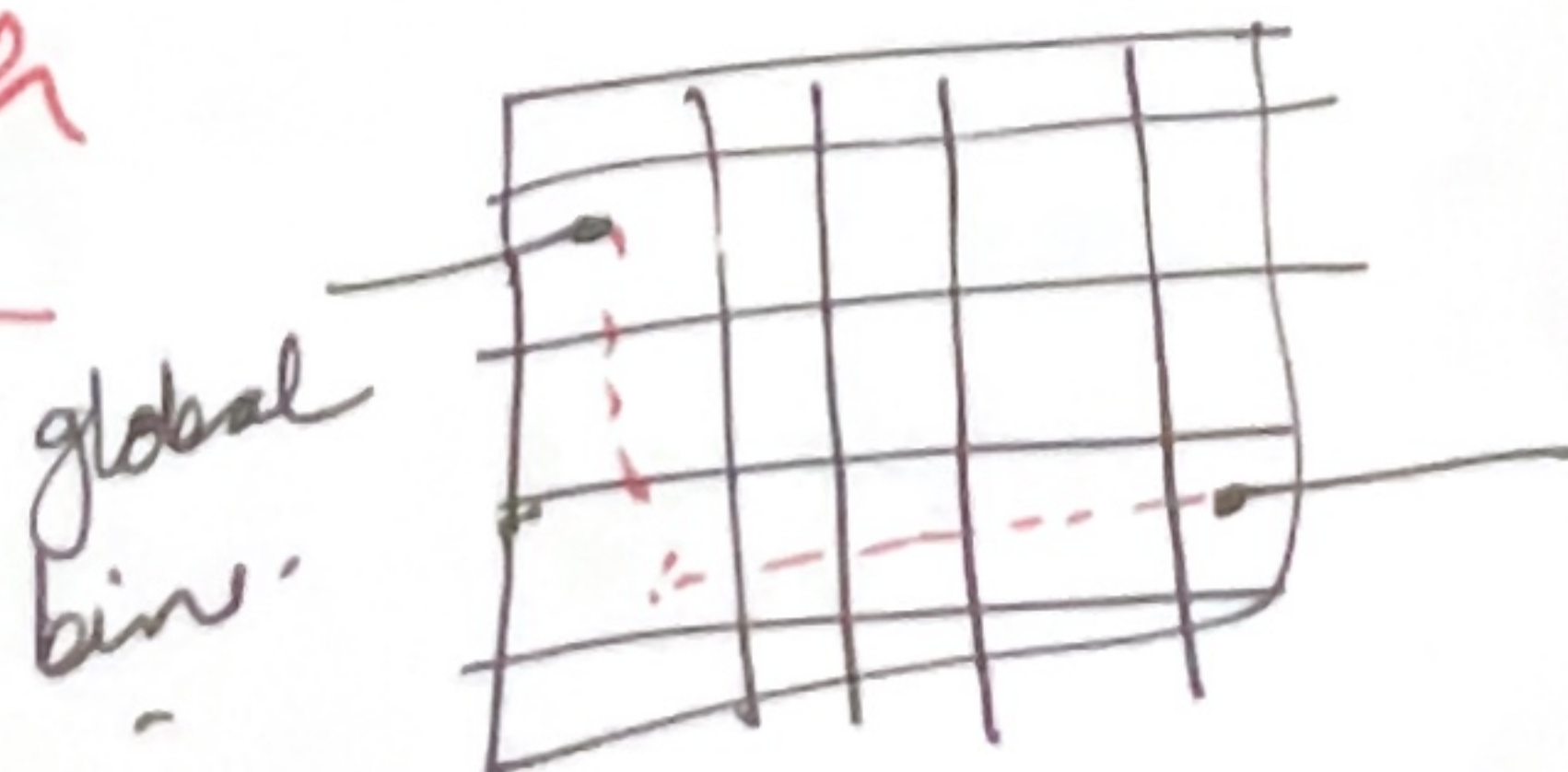
- The minimization of power dissipation

- Must consider this dissipation

→ Using Clock gating

Routing → creates wire ~~net~~ layout for all the nets, other

Than clk (To minimize wire length and power supply)



- Minimize Delay.

→ Routing - Global - Not creating actual wired layout.

- A Routing Plan for a given net.

- Detailed Routing. → Actual layout of each net in the for assigned global bins.

Metadata examples.

Flow plan

ECO, engineering change order

→ small final fixes.

→ write GDS.

↳ features of masks.

Overview of VLSI Design flow - V

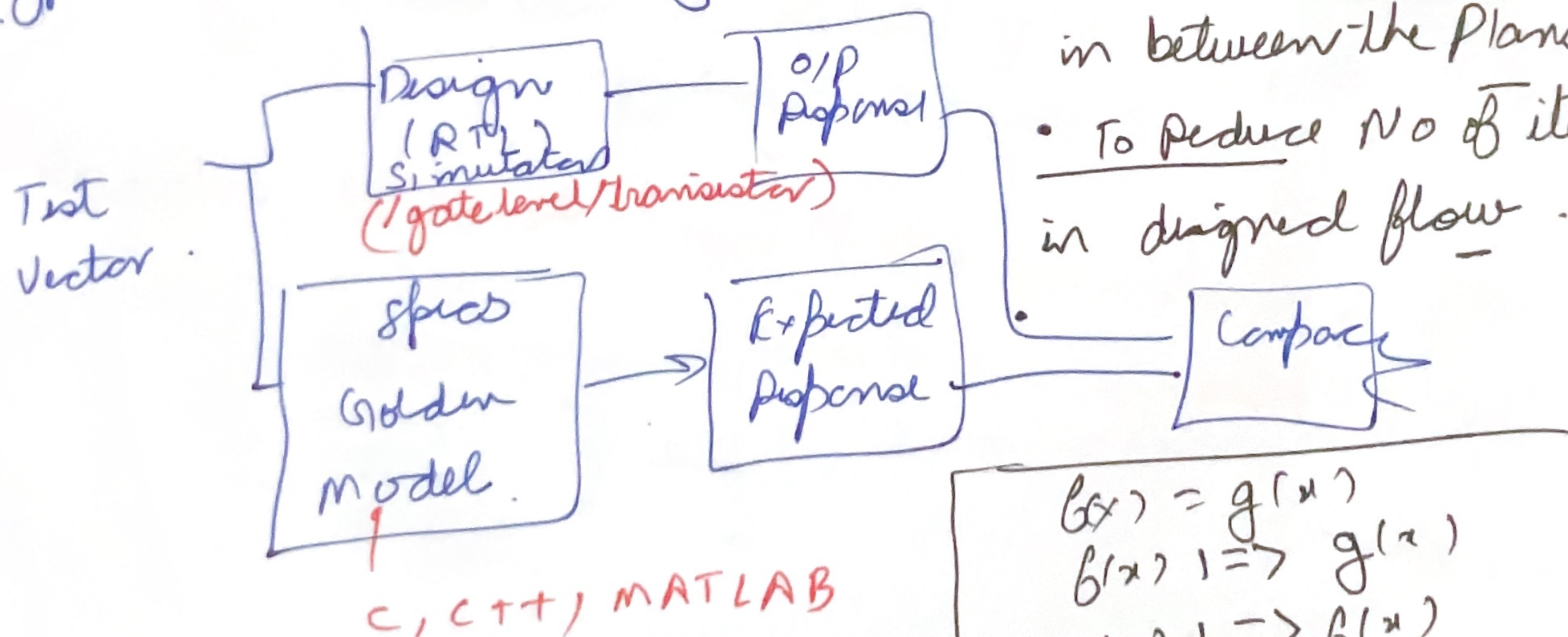
→ Verification, Testing.

human error,

Verification - design works as per functionality.
- Multiple times done in flow.

- immediate action.

Types - Simulation (using test vectors).



→ Test vectors are Not exhaustive
Model checking.

using formal methods

Establish the proof of a given property using formal mathematical tools such as deductions.

- covers all scenarios.

Functional Verification.

C E C - will always take place, b/w stages, we are sure that functionality is equal to RTL.

elements

Changes, insert buffers.

in Routing Placement and all.

Refinements, not large b/w each process of the flow. (Targetted)

wire length → Timing
one particular net cannot be routed for a particular net.

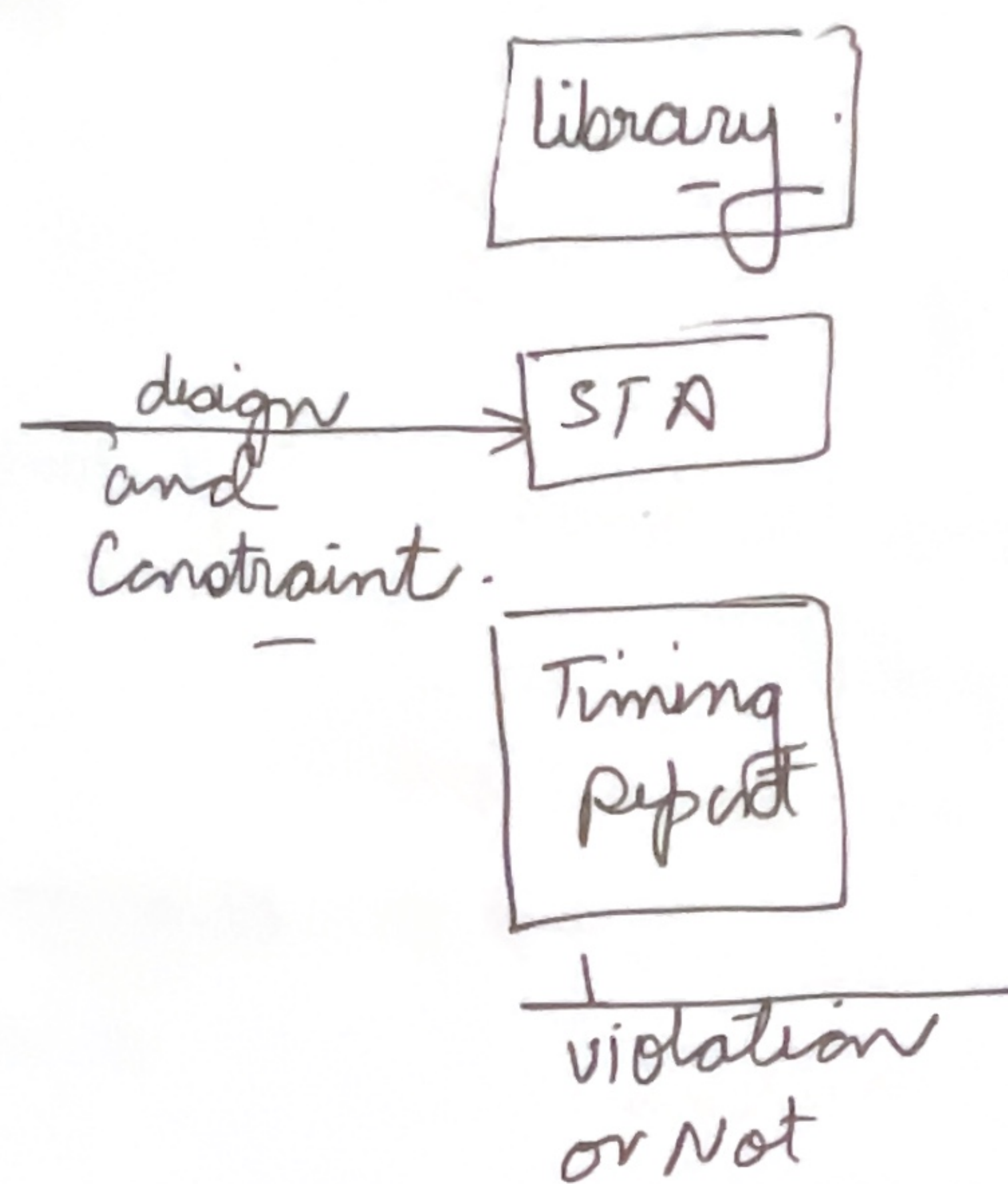
There can be loops in between the plans.

To reduce No of iterations in designed flow.

$$\begin{aligned} b(x) &= g(x) \\ b(x) &\Rightarrow g(x) \\ g(x) &\Rightarrow b(x) \end{aligned}$$

Computational difficulty

Prefer ? when High impedance is don't care.
STA → Considers Worst Case Behaviour.



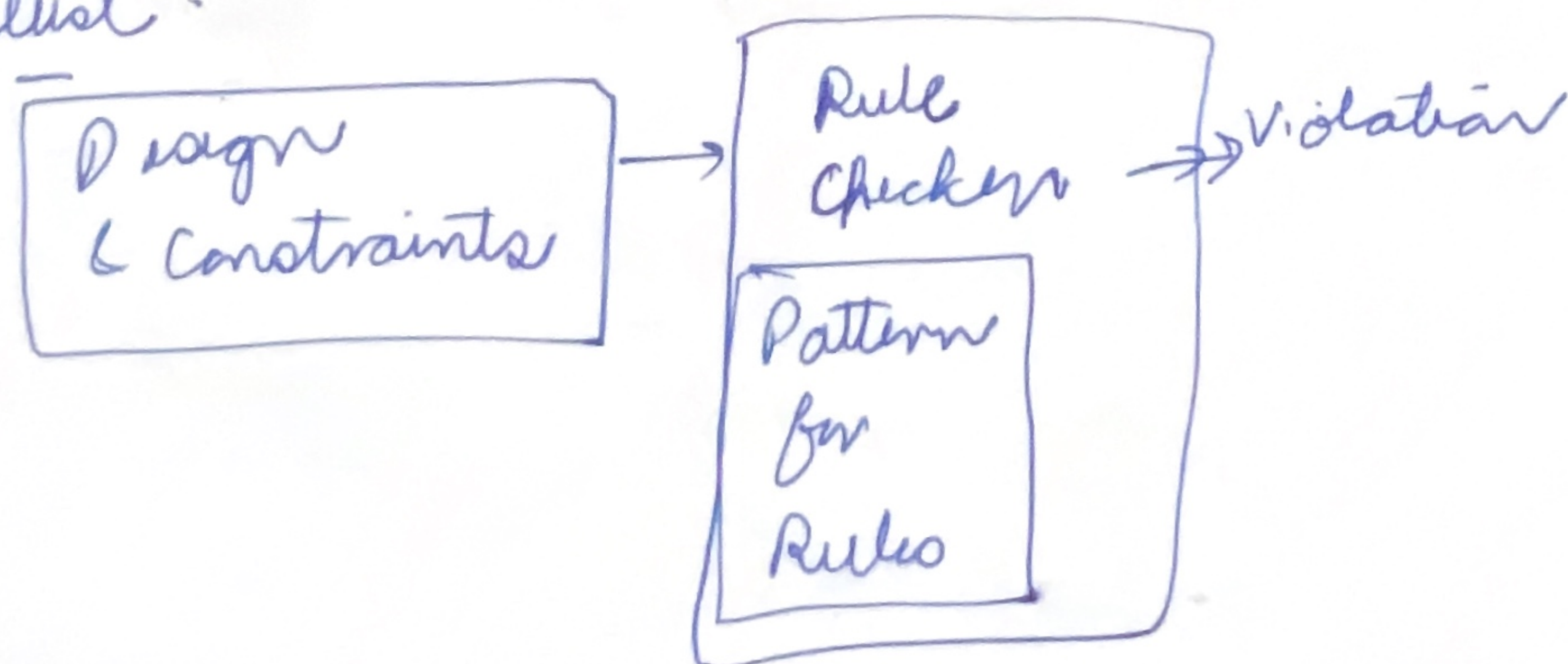
→ Physical Design Verification.
① Physical verification → To check during physical design.

② Design Rule Check (DRC)
— Defined by foundries, technology dependent.

→
③ E R C Electrical Rule Check — Connectivity.

④ Layout vs schematic (LVS) check.

Ensure that the layout is functionally eq To netlist.



FCO. Manufacturing

Testing - fab chips does not have any manufacturing defects.
- meeting original design.

- We have to add features, so it can be tested easily.

Defects: Physical imperfections in physical chip, that are permanent nature.

fabrication → clean Room.

Temperature Variance → misalignment of mask,

- Area defects can arise.

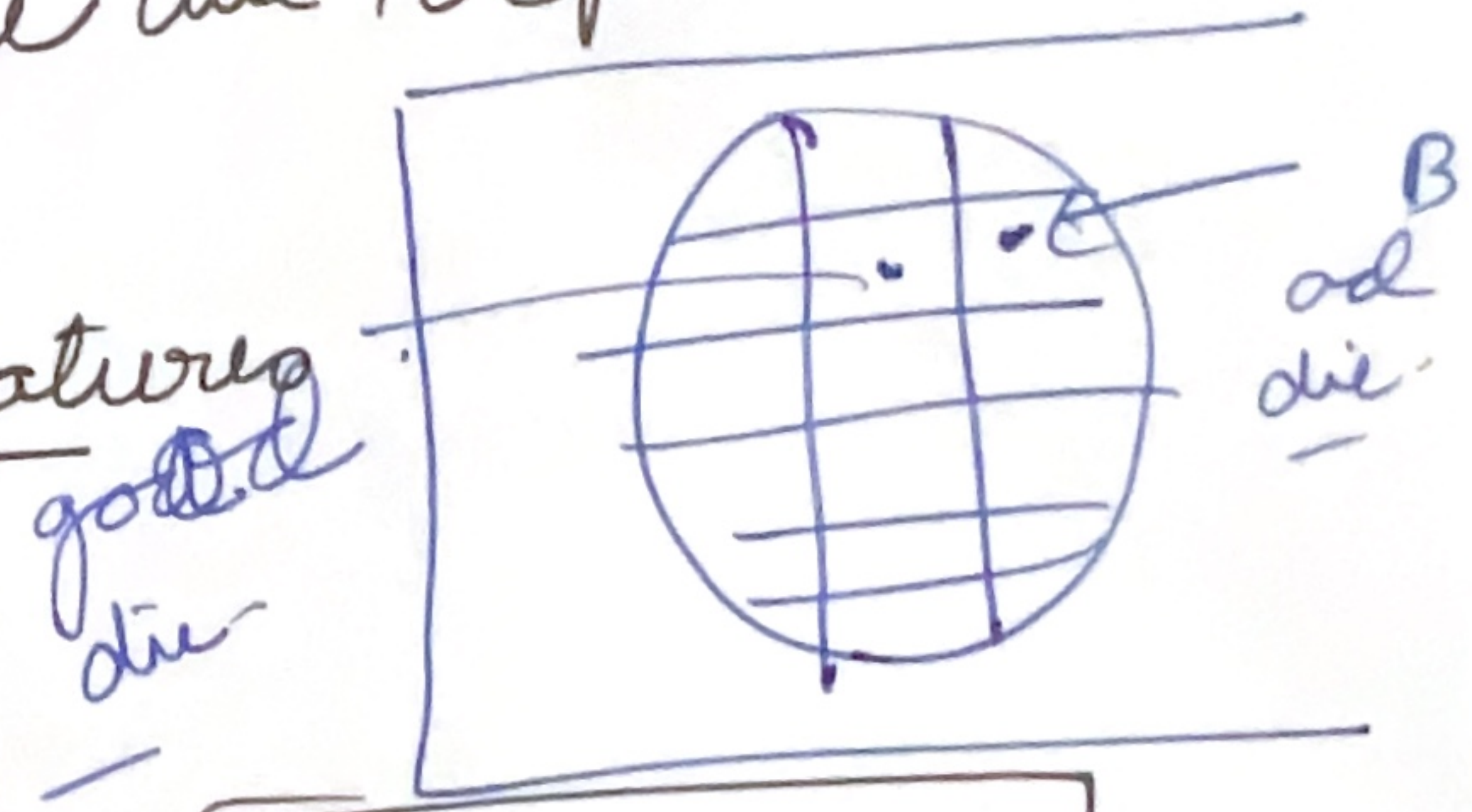
manufacturing defects.

SOC / O.C of lines -

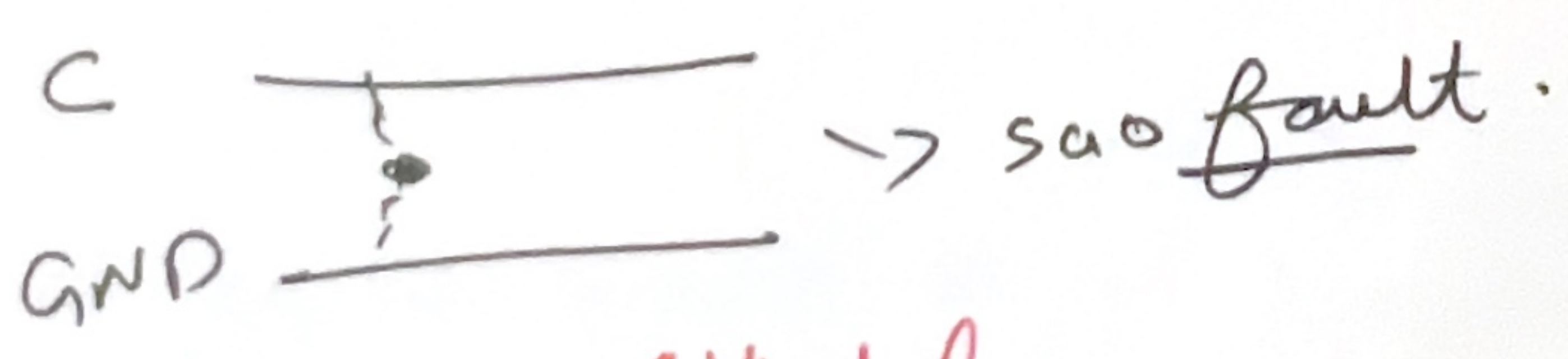
Distortions - Photolithography can produce distorted features on a die.

- These are inevitable due to optical effects etc.

→ Distorted features
→ good die



line C is SAC



Testing to detect that fault.

→ Quality of Process: Yield.

→ Fraction of die on a wafer that are good

Yield, 400, 300 (good).
 $300/400 \times 100 = 75\%$ yield.

Desirable > 50%.

- Die area, Defect P ↑ probability, yield ↓.

Uncollected -

clustered defect $\rightarrow$ non yield. (usually clustered in fabrication)

Yield Model

$$Y = 1 + \left(\underbrace{A d}_{\text{Area}} / \underbrace{\alpha}_{\substack{\text{cluster} \\ \text{Parameter}}} \right)^{-\alpha} \times 100$$

defect density

$$A d = 1,$$

α - low,

d = defect density.

Testing (ATE) Automatic testing equipment.
 $\rightarrow$ Load Test Patterns,
Observe Response.

Fault Coverage = No. of measure of the ability of the set of test patterns to detect a class of faults.

100 chips, yield = 90%, - 10 (bad)
- 90 (good)

if fault coverage = 50%.

5 bad chips may get undetected

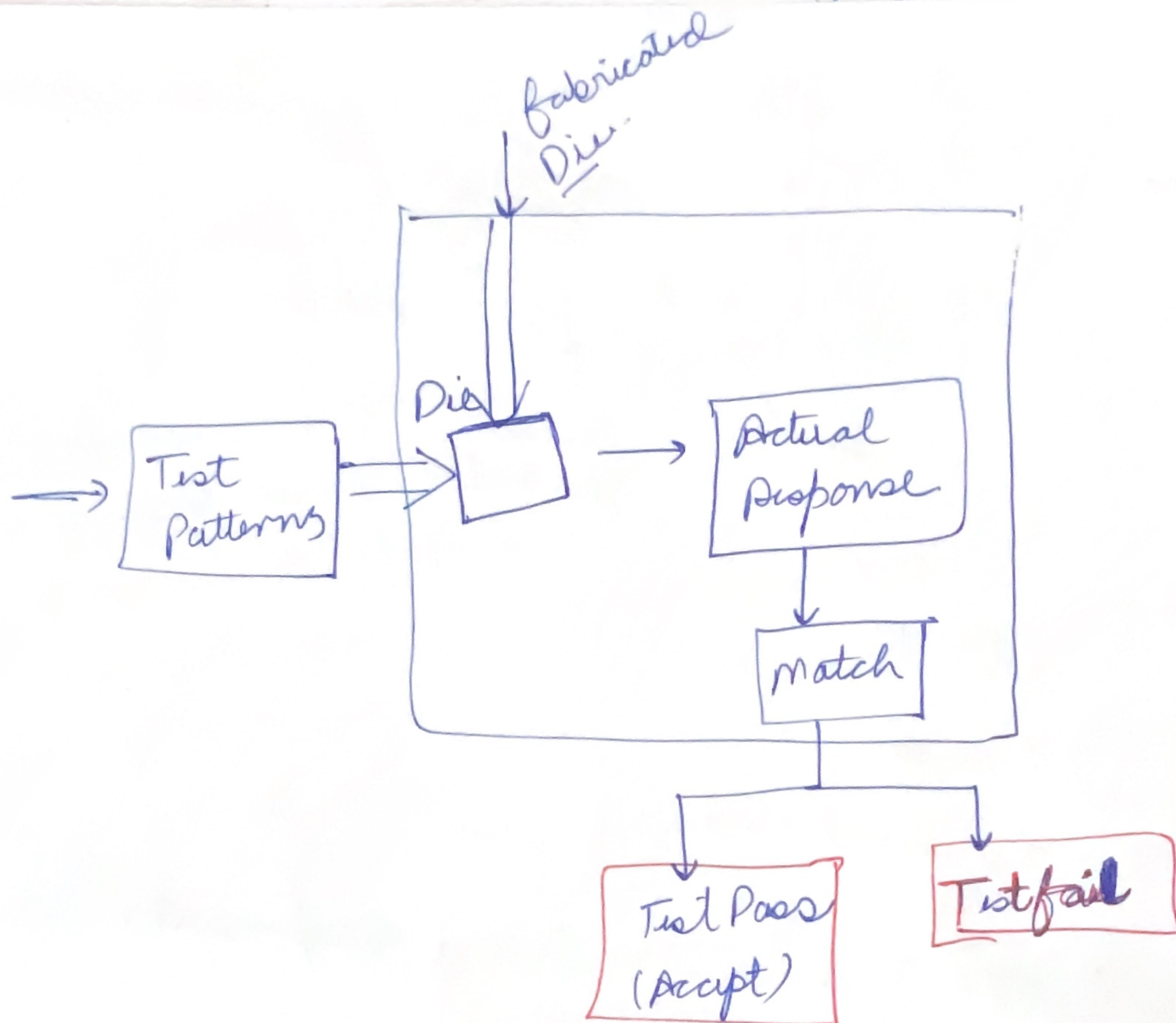


95 chips reach end customer

DL is a measure of effectiveness of the test

$$\frac{5}{95} \times 10^6$$

$\leftarrow$



→ Fabrication.

→ Mask fabrication is a replica of the patterns on a given layer of the layout created on the substrate (glass), for transferring patterns.

- Corresponding IC, fabrication.

- Data preparation

- Mask writing

- Quality checks and adding protections

PPT screenshot
Photolithography

